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基于SIW的介电系数宽带测量装置

陈虎¹, 陈倩², 刘长军², 黄卡玛², 龙卓²(¹ 成都航空职业技术学院信息工程学院, 四川 成都 610100; ² 四川大学电子信息学院, 四川 成都 610064)

摘要: 自20世纪80年代以来, 微波能的应用几乎扩展到了化学、材料、医学等各个领域。而微波能的应用实际上都直接或间接地与物质的介电特性相关。通过了解物质介电特性, 就能研究它们对于微波的吸收和反射情况。因此, 微波能应用中的介电系数测量显得尤为重要。而传统的测量方法实现在线测量时有一定的局限性, 针对上述问题设计了新型的测量装置, 结合神经网络算法对物质的介电系数进行了仿真计算, 得到了一些有参考价值的数据, 这些工作将对微波能的应用提供帮助。

关键词: SIW; 测量; 介电系数; 神经网络; 模拟

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Broadband permittivity measurement apparatus using substrate integrated waveguide structure

CHEN Hu¹, CHEN Qian², LIU Changjun², HUANG Kama², LONG Zhuo²¹ College of Electronical and Information Engineering, Chengdu Aeronautic Polytechnic, Chengdu 610100, Sichuan, China;² School of Electronics and Information Engineering, Sichuan University, Chengdu 610064, Sichuan, China)

Abstract: The microwave energy has been applied to chemistry, materials, and medicine since 1980s. The application of microwave energy is related with permittivity of materials. If the permittivity of materials is known, the absorption and reflection on microwave energy can be obtained. Hence the measurement for permittivity in microwave power applications is very important. However, the traditional measurements have some limitations on realtime measurement. The measurement apparatus has been proposed and the permittivity of materials has been reconstructed by modern optimization algorithm (artificial neural network). The applications of microwave energy have been extended by the valuable data gained. The research work will contribute to application of microwave power.

Key words: substrate integrated waveguide; measurement; permittivity; neural network; simulation

引言

微波技术经过几十年的发展, 已成为一门比较成熟的学科, 在雷达、通信、电子对抗、工业生产及科学研究等方面得到广泛应用。自20世纪80年

代以来, 微波能的应用几乎扩展到了化学、材料、医学等各个领域^[1-8]。而微波能的应用实际上都直接或间接地与物质的介电特性相关。因为物质的介电特性反映了其对微波的吸收和反射情况, 所以研究物质与微波的相互作用中一个重要而基础的问

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通信作者: 陈倩(1975—), 女, 博士, 副教授, chenqian@scu.edu.cn

第一作者: 陈虎(1973—), 男, 硕士研究生, 副教授, novman@163.com

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题就是研究其介电特性。因此在微波能的应用中测量物质的介电特性具有重要意义。

测量物质介电系数的方法有很多种,每一种方法都有它的优势与局限性。初略地可以将微波测量介电系数的方法分为谐振法和非谐振法,谐振法比非谐振法具有更高的测量精度和灵敏度,但测量带宽窄^[9-15]。而非谐振方法适用于在较宽的频带范围内对物质的介电系数进行测量,同时与谐振法相比,非谐振法一般不需要样品的准备,测量过程也比谐振法更简便,使得测试更易于实现^[16-20]。近来SIW(substrate integrated waveguide)这种新型的波导结构已经用于测量装置的设计中,已有文献报道基于谐振法的SIW测量装置,在频率为5.8 GHz时取得了准确的测量结果^[21]。与传统的同轴结构宽带测量装置相比,SIW结构简单,易于加工,实测模型更接近仿真模型,便于提高测量精度^[22-26]。

近来在反演物质介电系数的应用中,人工神经网络计算模型作为一种非传统的有效方法已逐渐得到人们的认可,神经网络通过学习微波辐照下测试系统中物质等效介电系数与散射参量之间的关系,可以快速、准确地得到待测物质的介电系数^[27-29]。因此,为在线测量物质介电系数提供了一个有效途径。

本文对基于SIW结构的宽带介电系数测量方法进行的研究,设计了相应的测量装置,能实现物质介电系数的宽带测量,仿真及实测结果表明SIW结构能用于介电系数宽带测量。

1 测量装置的设计

本文基于传输反射法的原理,分析SIW结构的特点,研究电磁波在SIW结构中的传输机理,设计了基于SIW的宽带介电系数测量结构。

1.1 基于SIW结构的测量装置

首先设计基于SIW的宽带介电系数测量结构,仿真验证结构合理性。在SIW结构传输机理研究的基础上设计灵敏度较高的介电系数测量结构。使测量结构散射参量的变化能反映待测物介电系数的变化。仿真测量装置模型如图1所示。测量装置分为三个部分,分别为50 Ω 微带线、微带过渡段和SIW测量段。测量段为中间开缝的SIW结构,测量装置实物照片如图2所示。实际测量时,测量装置通过电缆与矢量网络分析仪连接构成测量系统。

测量装置水平放置,将不同待测物放置于开缝处上方,改变不同待测物,即改变待测物质介电系数的实部和损耗角正切,观察所得散射参量的变化情况。在SIW装置的测量缝周围采用热熔胶将待测液体固定在测量缝上方。

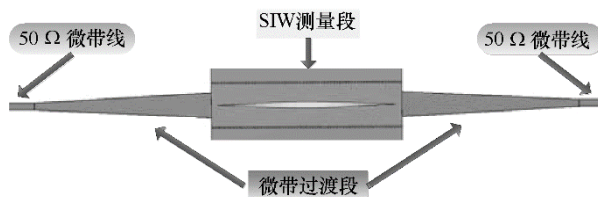


图1 测量装置仿真模型

Fig.1 Simulation model of measurement apparatus

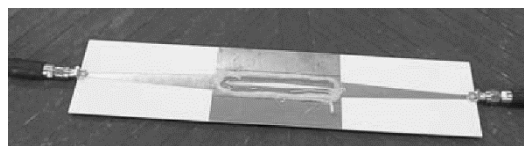


图2 SIW测量装置照片

Fig.2 Photo of measurement apparatus

1.2 实验及仿真结果

为仿真验证结构设计的合理性,本文采用了全波电磁仿真软件对所设计的装置模型进行了仿真计算,在仿真计算的基础上,加工测量装置,搭建实验测试系统进行实测。在仿真计算及测量装置加工时,基板材质选取了Rogers RO4003C,介电系数实部为3.38,损耗角正切为0.0027,基板厚度为0.813 mm,仿真计算及实验测试结果如图3~图5所示。

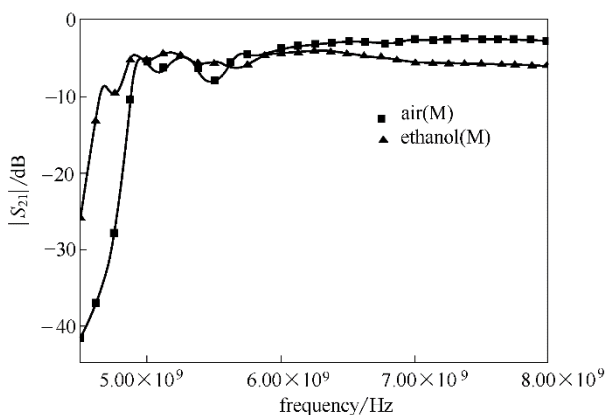


图3 空气及酒精实测 $|S_{21}|$

Fig.3 Measurement magnitude of transmission coefficient

图3为待测物为空气和酒精时,散射参量 S_{21} 幅度 $|S_{21}|$ 的对比曲线,图4为待测物为空气和酒精时散射参数 S_{21} 相位 $\Phi_{S_{21}}$ 的对比曲线,图3、图4表明 $|S_{21}|$ 和 $\Phi_{S_{21}}$ 均能在频率范围为5.5~8 GHz情况下,明显区

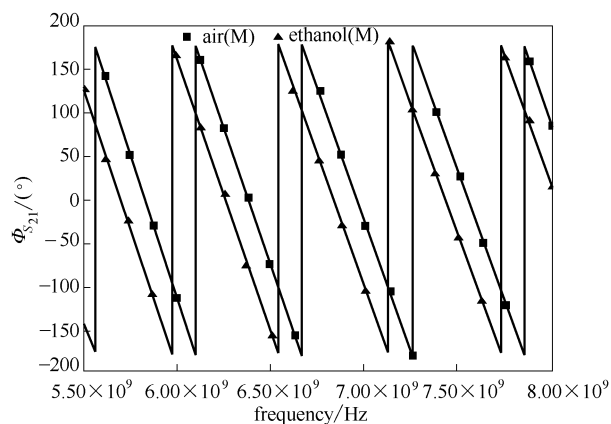
图4 空气及酒精实测 $\Phi_{S_{21}}$

Fig.4 Measurement phase of transmission coefficient

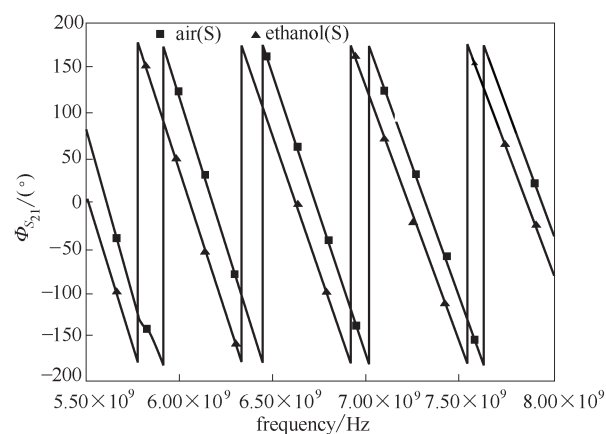
图5 空气及酒精仿真计算 $\Phi_{S_{21}}$

Fig.5 Simulation phase of transmission coefficient

分酒精和空气两种物质。图5为待测物为空气和酒精时, S_{21} 相位 $\Phi_{S_{21}}$ 的仿真计算结果, 仿真结果图5与实测结果图4变化趋势一致。在同一频点, 待测物为空气或酒精时, 两种物质 $\Phi_{S_{21}}$ 相位差的仿真计算值与实测值是一致的。仿真和实测数据表明在实测时可以用空气作为校准物进行测量, 以提高测量精度, 同时也为后续利用仿真数据训练神经网络反演待测物介电特性提供了基础。

2 反演算法

2.1 基于人工神经网络的反演算法

基于人工神经网络的反演算法在介电系数测量中已经得到应用。在仿真SIW测量结构的基础上, 得到大量介电系数实部、损耗角正切及相应的散射参量的数据, 将仿真所得的介电系数实部、损耗角正切及散射参量作为神经网络的训练样本, 以

散射参量为神经网络输入, 物质介电系数实部和损耗角正切为输出, 构建人工神经网络反演模型, 优化人工神经网络反演模型, 当介电系数重构精度达到要求时, 将所得神经网络模型存储, 最终就可以将仿真或校准后实测的散射参量作为神经网络的输入, 神经网络的输出为待测物质的介电系数实部与损耗角正切, 实现物质介电特性的在线测量。

2.2 反演结果

本文将频率5.8 GHz酒精作为待测物1, 标准值为文献[30–31]中查找数据, 反演值为代入实测散射参量数据计算得到。假定待测物2为仿真的散射参量加2%误差作为实测数据, 代入训练好的神经网络中, 所得结果如表1、表2所示。反演结果表明, 对于酒精和假定标准物质2, 该测量装置所测介电系数的实部和损耗角正切误差都在5%以内, 具有较高的测量精度。此外, 该装置还可以用于小于8 GHz其他频点介电系数的测量。

表1 介电系数实部仿真反演

Table 1 Inversed real part of complex permittivity

待测物质	介电系数的实部		
	反演值	标准值	相对误差/%
待测物1	4.65	4.85	-4.1
待测物2	11.62	12	-3.2

表2 介电系数损耗角正切仿真反演

Table 2 Inversed loss tangent of complex permittivity

待测物质	介电系数的损耗角正切		
	反演值	标准值	相对误差/%
待测物1	0.67	0.65	3.1
待测物2	0.29	0.3	-3.3

3 结 论

本文依据传输反射法设计了基于SIW结构的介电系数测量结构, 为验证测量结构的有效性, 对比了5.5~8 GHz频带范围内, 待测物为酒精和空气时的散射参量, 表明所设计的介电系数测量装置具有较高的灵敏度, 同时仿真结果与实测结果的一致性, 为后续将SIW介电系数测量结构仿真所得数据作为训练样本, 构建神经网络结构, 反演待测物质的介电系数实部与损耗角正切提供了依据。最后验证了测量装置的合理性及测量方案的可行性。仿真及实测结果表明该装置具有较高的灵敏度, 测量结果准确, 能用于物质介电系数测量。同时, 该测量结构具有体积小、易加工的优势。

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